Arith Maldonado Zamudio

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GPA: 9.1/10.0 | Project portfolio: aritmaldzamu.github.io/xgio-monorep/descargas/

Professional Summary

Mechatronics Engineering student and Biomedical Engineer seeking an Automation Intern role in the automotive industry. Hands-on experience in medical equipment service, mechanical design, FEA, embedded prototyping, computer vision, and GPS-based software systems; skilled in building functional prototypes and solving technical problems across hardware, software, and manufacturing.

Education

B.S. in Mechatronics Engineering (In progress) — *Expected Dec. 2026*
Universidad Iberoamericana Puebla

- **GPA:** 9.1/10.0. **Relevant coursework:** Strength of Materials, Advanced Control Systems, Embedded Systems, Mechanical Design, Industrial Automation.

B.S. in Biomedical Engineering (Graduated with Honorable Mention) — *2020 - 2024*
Universidad Iberoamericana Puebla

- **GPA:** 9.1/10.0. Focus in biomechanics, medical device design, finite element analysis, and prototyping.

Relevant Experience

Technical Specialist & Medical Equipment Application Representative — *Jul. 2024 - Dec. 2024*
Punto Focal Equipo Médico, Puebla, Mexico

- Performed preventive/corrective maintenance, installation, and troubleshooting of ultrasound devices, X-ray systems, flat panels, and triggers.
- Supported 15+ ultrasound-related service cases, preparing technical reports, client follow-up, warranty/ticket documentation, and basic diagnostics.
- Delivered technical training and product demonstrations to physicians, technicians, and customers.
- Identified a functional replacement for a medical power supply, reducing client cost from approx. MXN $15,000 to MXN $1,500.

Selected Projects

3-DOF Ball-Balancing Platform — *Mechatronics*
ESP32, SG90 servos, Python, OpenCV, PID control, Bluetooth communication

- Built a functional closed-loop mechatronic platform that uses camera-based feedback and continuous PID control to balance a ball at the center of the plate.
- Integrated embedded control, servo actuation, real-time vision processing, and PC-to-microcontroller communication for automation-focused prototyping.

XGIO Smart Cane GPS Tracking Ecosystem — *Social Service*
Mobile app, backend, admin dashboard, Firebase, Vercel, CAD/SolidWorks documentation

- Developed a functional GPS-based system for people with visual impairment, including app, backend, administrative dashboard, user-device registration workflows, and technical documentation.

Honors, Certifications and Skills

- **Honors:** Poster accepted at ISPO 20th World Congress 2025, Stockholm: “Redesign and validation of a tool holding system for a transradial prosthesis for a stomatology student”; Honorable Mention in Biomedical Engineering, Dec. 2024.
- **Certifications:** CSWA, Certified SOLIDWORKS Simulation Associate, Certified SOLIDWORKS Additive Manufacturing Associate, Google Project Management Certificate, EF SET English B2.
- **Technical skills:** SolidWorks, SolidWorks Simulation, MATLAB, Python, OpenCV, Arduino, ESP32, Firebase, Altium Designer, Power BI, Excel, CNC/router CNC, lathe, milling, laser cutting, 3D printing, drawing interpretation, and metrology.